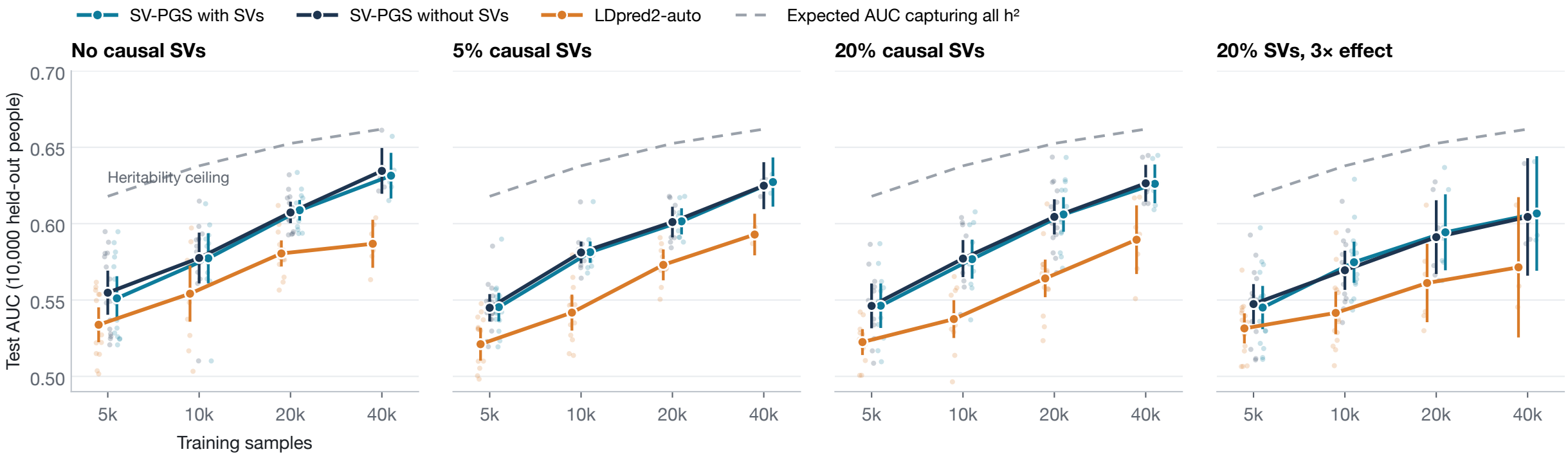
SV-PGS gains accuracy faster than LDpred2-auto as training grows

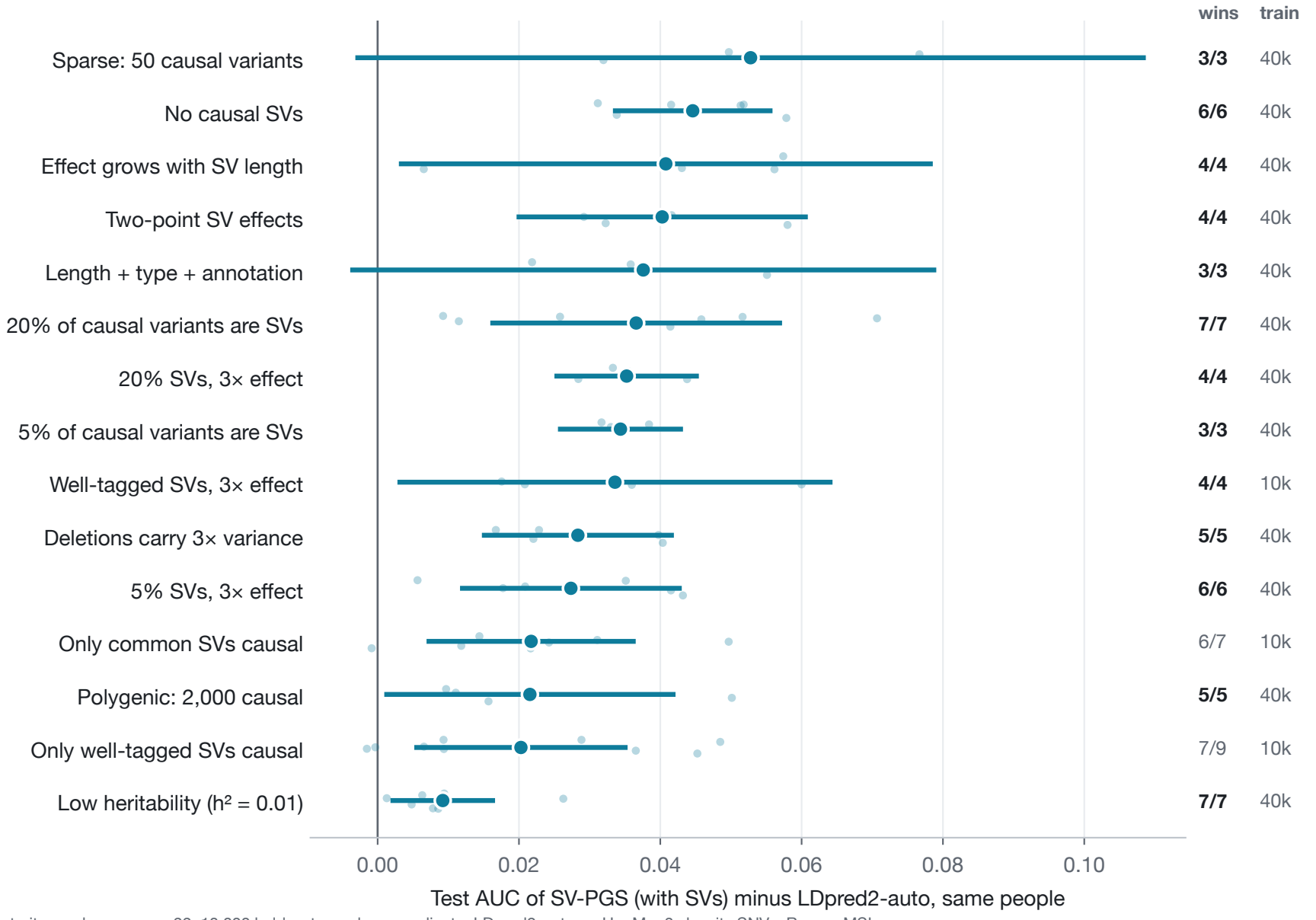
Simulated binary trait on chromosome 22 ($h^2 = 0.1$, prevalence 10%, 200 causal variants); means with 95% intervals, faint dots are replicates



bench-sim v7 cohort (1000 Genomes haplotypes), Beagle-imputed SVs; LDpred2-auto fitted on HapMap3-density SNVs of the same training people. Run on MSI.

SV-PGS beats LDpred2-auto in 74 of 77 paired replicates, across 15 architectures

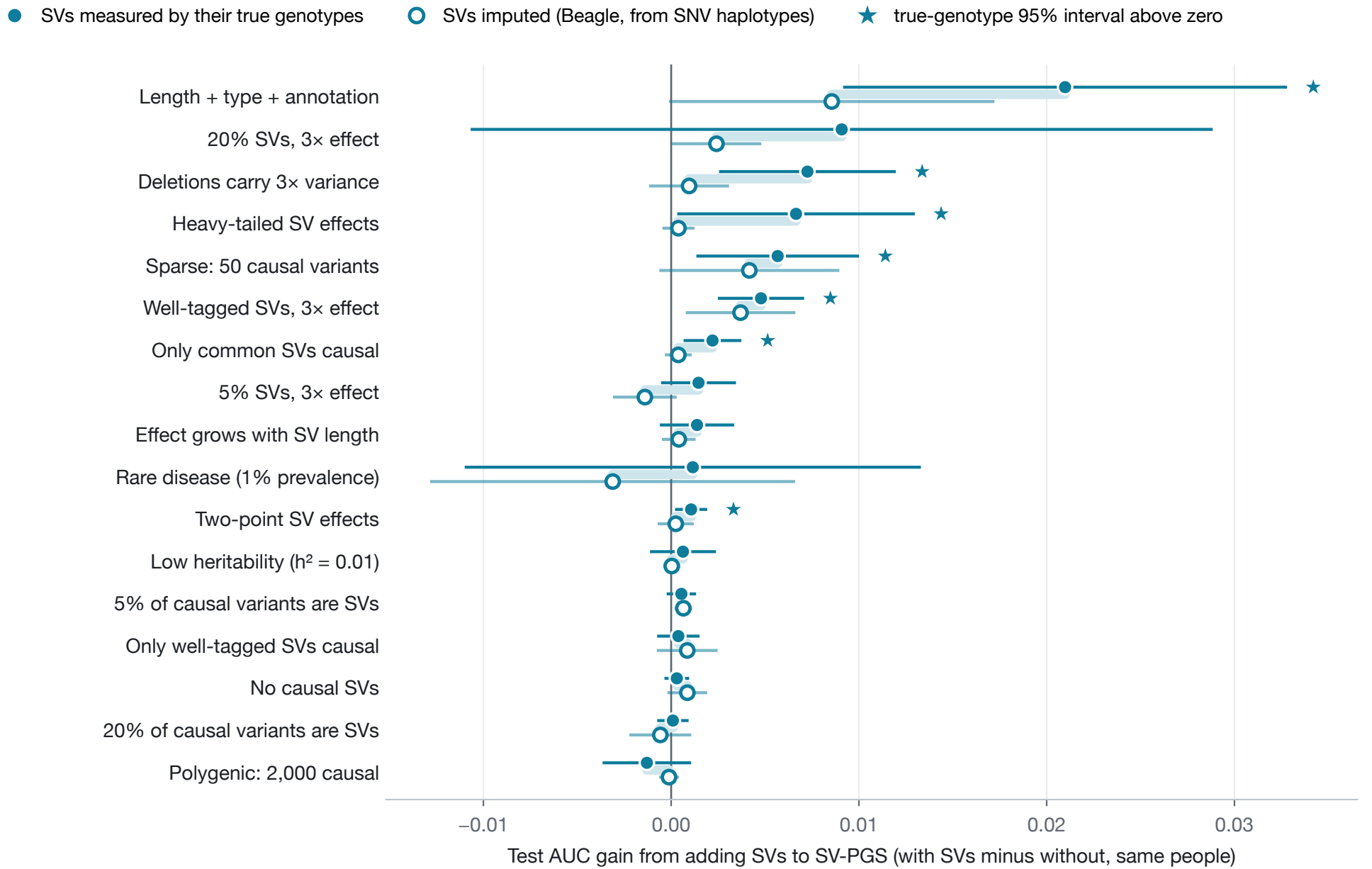
Paired difference in held-out AUC per replicate (dots), mean and 95% interval, at the largest training size with ≥ 3 replicates



Simulated binary traits on chromosome 22; 10,000 held-out people per replicate. LDpred2-auto on HapMap3-density SNVs. Run on MSI.

SVs add accuracy when they are measured well; imputation erases most of the gain

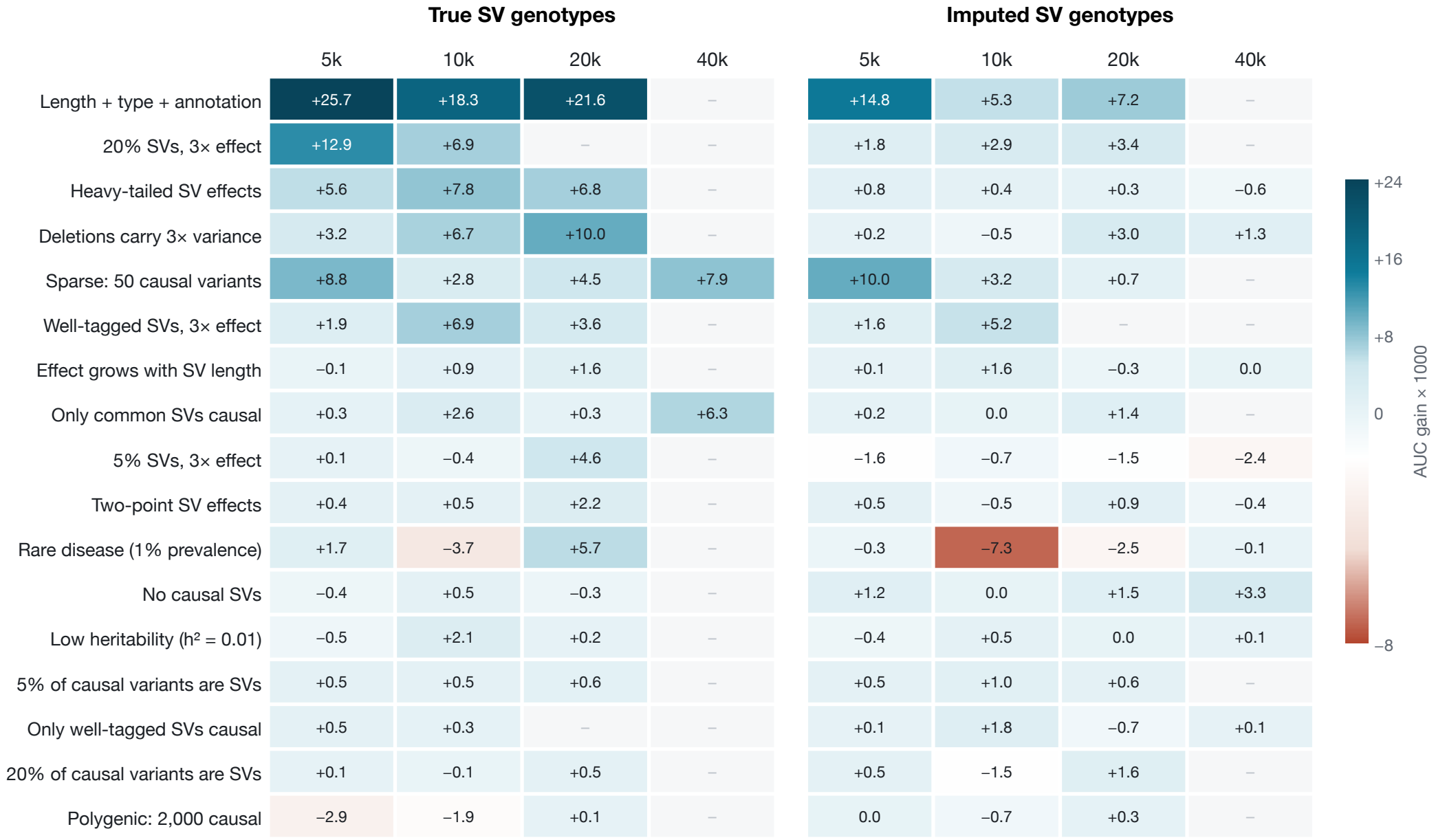
One 3 Mb chr22 window (60,000 records), every architecture and training size (5k–40k) pooled; means with 95% intervals over replicates



Same simulated people, truth and fits; only the SV genotypes differ. Run on MSI.

Where adding SVs helps: architecture × training size

Mean held-out AUC gain from adding SVs (×1000) by training samples (columns), one chr22 window; cells with fewer than two replicates are blank



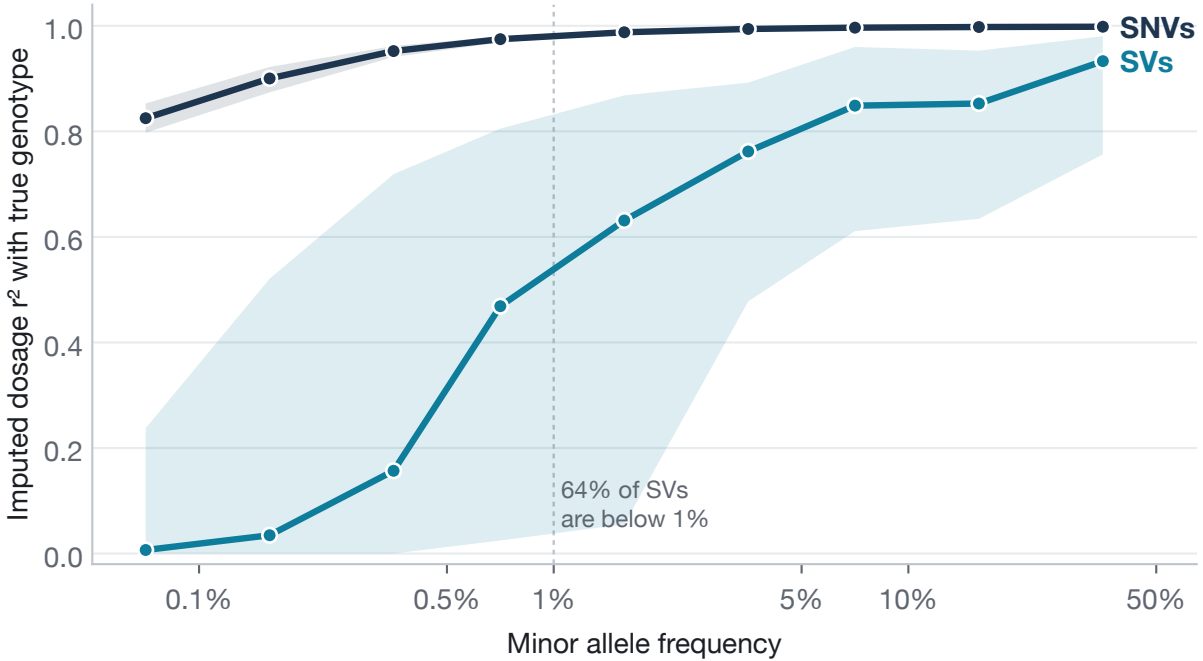
Simulated binary traits; rows ordered by the true-genotype gain. Run on MSI.

The simulated SVs: mostly rare, short, and hard to impute

Beagle imputation of 50,000 people from 1000 Genomes haplotypes on chr22: 1,662 SVs and 6,000 sampled SNVs scored against their true genotypes

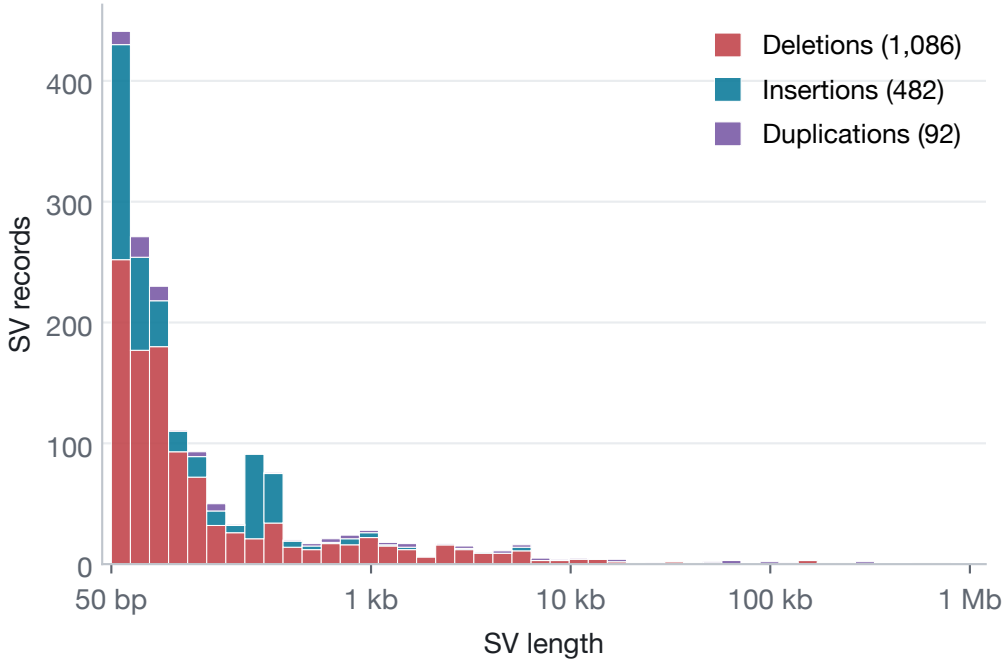
A Rare SVs impute poorly

median and interquartile range per frequency bin



B The chr22 SV catalogue

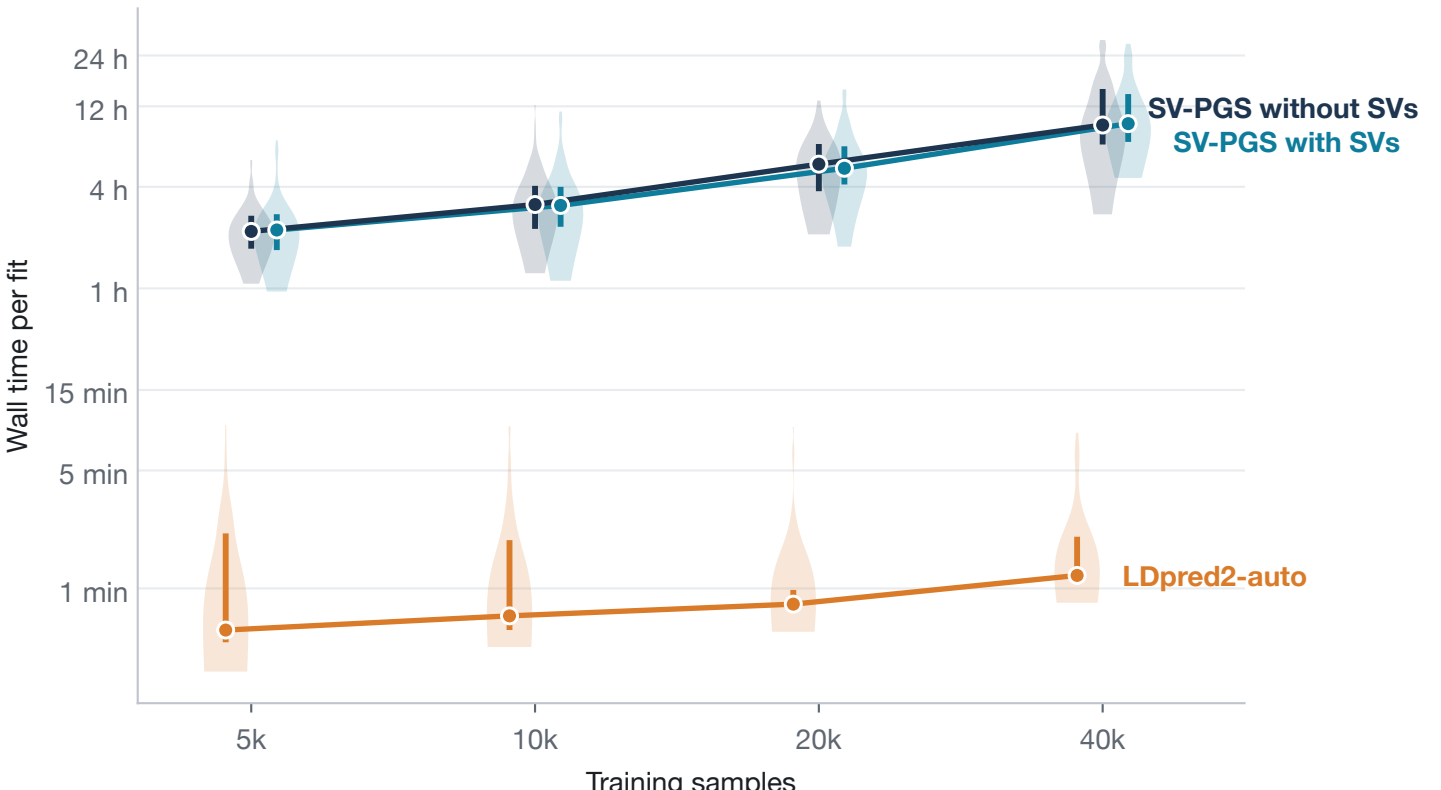
1,662 measured SV records, stacked by class



bench-sim v7 cohort (public 1000 Genomes data). Run on MSI.

What the accuracy costs: fit time per replicate

Whole chromosome 22; median, interquartile range and distribution over every architecture



MSI Slurm jobs, one fit per job. SV-PGS fits every variant by empirical Bayes; LDpred2-auto fits HapMap3-density SNVs.